

CoBit on Sudoku: bitstream diffusion vs. S-FLM baselines

Continuous bitstream diffusion (CoBit) evaluated under the S-FLM verifiable-task protocol.

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Task. We replicate the Sudoku benchmark from *S-FLM* (Deschenaux & Gulcehre) under a fair, ground-truth protocol. Each example is a 9×9 puzzle laid out as a 180-token sequence [BOS] puzzle(89) [BOS] solution(89) over a 12-id vocabulary. The first 91 tokens (the puzzle prompt) are conditioned and excluded from the loss; the model must generate the 89-token solution suffix. Difficulty is set by clue count — **easy/medium/hard** = 40/35/30 clues — with a 48k-train / 2k-val split (seed 42, deduplicated, unique-solution). The metric is *exact match*: the entire 89-token solution must be correct ((gen = gt).all). Vocab, layout, split, difficulty generator, and metric are byte-for-byte identical to S-FLM.

Method under test. CoBit replaces S-FLM's hyperspherical token-embedding state with a fixed-width *analog-bit* state (4 bits/token $\rightarrow$ 720-bit stream) and trains Gaussian bitstream diffusion (matched-filter residual, binary score matching, EDM weighting, entropy-rate noise schedule). The prompt is imposed by clamping the clean prompt bits at every solver step. Sampling uses the `ddim_entropic` path (probability-flow on the entropy-rate σ -grid with EDM stochastic churn $s_{\text{churn}} = \gamma(\text{NFE}-1)$), the same sampler that produces CoBit's OWT GenPPL numbers. Trunk is matched to S-FLM (8 blocks / 512-d / 8 heads, 20k steps, batch 256, AdamW lr 3×10^{-4} , NFE = 180).

Headline result. Sweeping the diffusion temperature γ (3 seeds, mean $\pm$ s.d.), CoBit **beats every published baseline on all three difficulties**, with the largest margin on *hard* (+7.5 pts over the strongest prior method):

Method (180 NFE)	easy	medium	hard
CoBit (best γ)	98.5	91.7	65.9
Duo	96.3	84.7	58.4
S-FLM (+trunc+adaptive)	94.8	85.2	45.0
FLM	94.2	82.7	44.5
AR (greedy)	14.6	5.1	1.0

Temperature sweep (CoBit exact-match %, mean over seeds 42/43/44). Stochastic churn helps *monotonically* and then plateaus near $\gamma \approx 0.4-0.5$, so the optimum is bracketed. On hard, churn lifts accuracy by +26 pts over the deterministic ($\gamma=0$) probability-flow ablation — the stochastic operating point is essential, not optional.

γ	0.0 (det.)	0.05	0.10	0.15	0.20	0.30	0.40	0.50
easy	93.3	96.4	97.3	97.7	98.0	98.4	98.5	98.4
medium	76.7	85.0	87.7	88.9	89.7	91.1	91.6	91.7
hard	40.2	53.5	58.6	61.4	63.0	64.6	65.9	65.8

Reporting note. Numbers use the raw (running) weights. The EMA weights are reported as collapsed: at only 20k steps the EMA decay 0.9999 (~ 10 k-step time constant) has not converged off the noisy early-training average, so EMA decoding falls out-of-vocabulary — expected at this step count, not a defect. Seed variance is tight (s.d. $\leq 1.3\%$).

Next: TinyGSM $\rightarrow$ GSM8K (prepared, ready to launch)

Task. The second S-FLM verifiable task: train on TinyGSM/TinyGSM (synthetic grade-school math with executable Python answers) and evaluate *executable-code accuracy* on the 1,319 GSM8K test problems. Each example is [BOS] question \n code [EOS], tokenized with SmoLM-135M ($|V| = 49152 + \text{PAD}$), length 512; the question is conditioned and masked out of the loss. Scoring runs the generated `simple_math_problem` in a restricted math-only sandbox (1e-3 tolerance, 5 s timeout) with a 95% bootstrap CI — the vendored S-FLM sandbox, unchanged.

CoBit setup. 16 bits/token $\rightarrow$ **8192-bit** bitstream; trunk matched to S-FLM (12 blocks / 768-d / 12 heads), 250k steps, global batch 512, EMA 0.9999, AdamW lr 3×10^{-4} . Eval at NFE = 1024 (the paper's headline budget) with a small γ sweep; bitstream decode that lands out-of-vocab (≥ 49153) is logged via an `invalid_token_rate` diagnostic.

Status & plan. Data pipeline, vendored GSM8K test set + sandbox, train/eval scripts, and config are in place and unit-tested; training has not yet started. We launch on Isambard (2 nodes $\times$ 4 GH200, DDP), then evaluate the 250k-step checkpoint.

Targets to match (GSM8K acc %, T=1). Non-AR diffusion band: MDLM 18.0, Duo 17.2, S-FLM(+adaptive+S-arch) 12.4, S-FLM(+top-k=1) 18.0, FLM 0.3; AR reference: greedy 63.3, sample 53.9. The fair first target for CoBit is the diffusion band ($\sim 17-18\%$), reporting at the same NFE as the method being compared.